BUS 9430 · CAPSTONE 3 · MODELS AND RESULTS

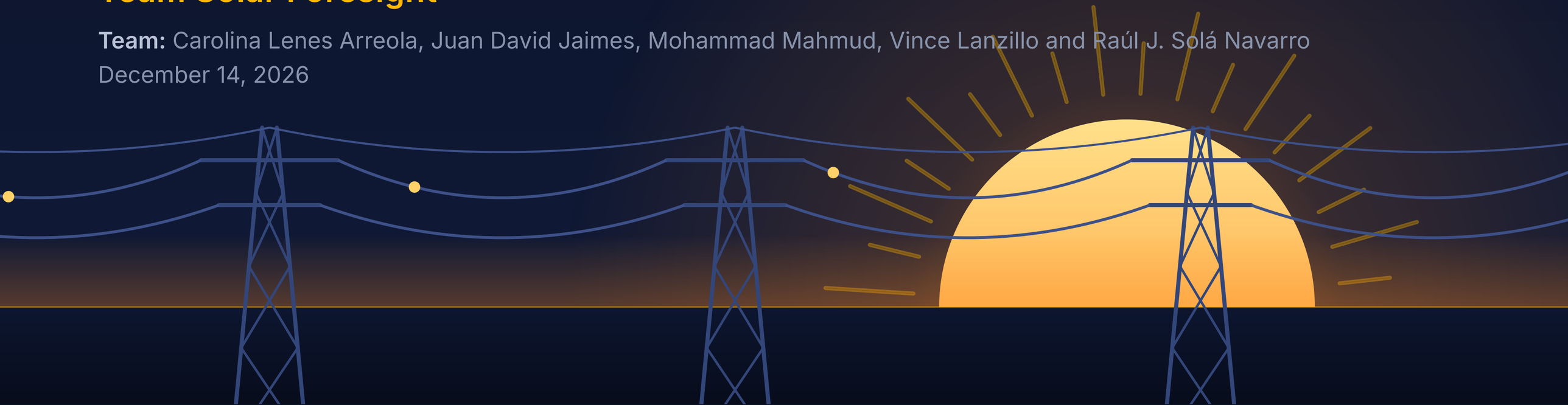
Tomorrow's Sun, Today's Plan

What we built, how well it works, and what it means

Team Solar Foresight

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THE PROBLEM AND FIVE QUESTIONS

Guessing tomorrow’s solar power, and five questions we tested

TARGET VARIABLE

Solar power for each hour of tomorrow

In megawatts (MW), for the whole CAISO grid.

THE RULE

Only what we know at 10 a.m.

CAISO's bids close at 10 a.m. Pacific, the day before.

1. Does the weather forecast help?
2. Can free public data get close to CAISO's own forecast?
3. Does a flexible model beat a straight-line one?
4. Do clouds make forecasts worse?
5. Does a better forecast mean less backup power?

We wrote the five questions and how we would judge them before we looked at the final test.

MODELING APPROACH

How we tested it, fairly

1

One table

The data from Capstone 2

22,896 hourly rows: weather forecasts, solar output, installed capacity.

2

Five forecasts

Scored on the same hours

Same as 2 days ago. CAISO's own. A monthly average day. A straight-line model. Our tree model.

3

Learn from the past

Earlier days only

Models learn from earlier days and are checked on four time-ordered blocks, never on days they trained on.

4

Lock, then test once

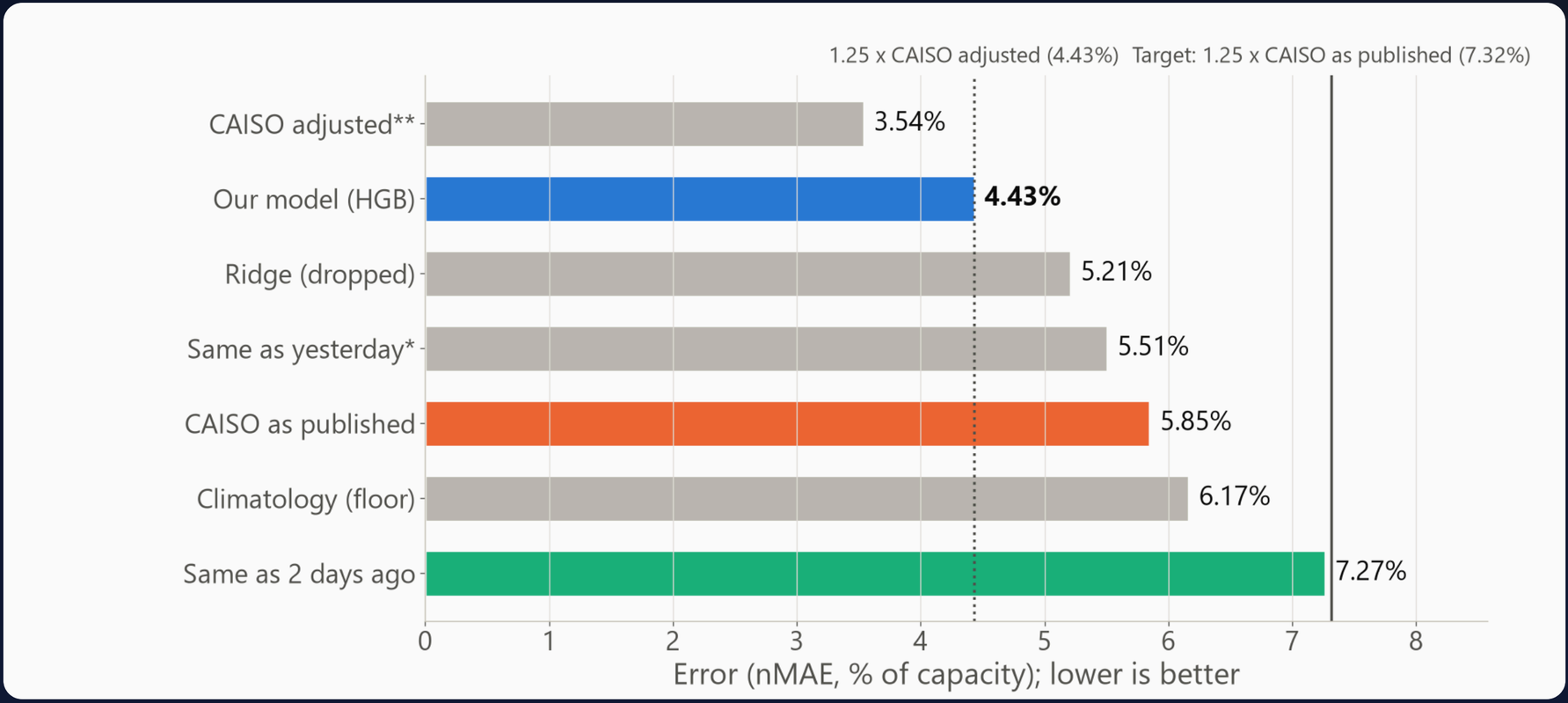
Last 191 days, Apr 10 to Oct 17

Settings and questions saved first. The test ran once. We report whatever came out.

Our model is a tree model: many small decision trees, each fixing the mistakes of the one before. It reads 20 inputs, all known at 10 a.m.: the 2-day-old weather forecast, time of day and season, sun height, installed capacity, and output from 2 days earlier.

RESULTS: THE SCOREBOARD

Our error is 4.4%, inside the 7.3% limit we set



Error is the average miss as a share of installed solar. **Ours 4.43%, CAISO as published 5.85%, same as 2 days ago 7.27%.** Some of CAISO's miss is solar turned down later. Adjusted, CAISO scores 3.54%, just under us.

Test set: 2,457 daylight hours, Apr 10 to Oct 17, 2026. Error = mean absolute error as a share of installed solar (EIA 860M capacity). * Same as yesterday uses output from after the 10 a.m. bid close, so no bidder could use it. ** CAISO adjusted is partly circular (see backup).

RESULTS: THE FIVE QUESTIONS

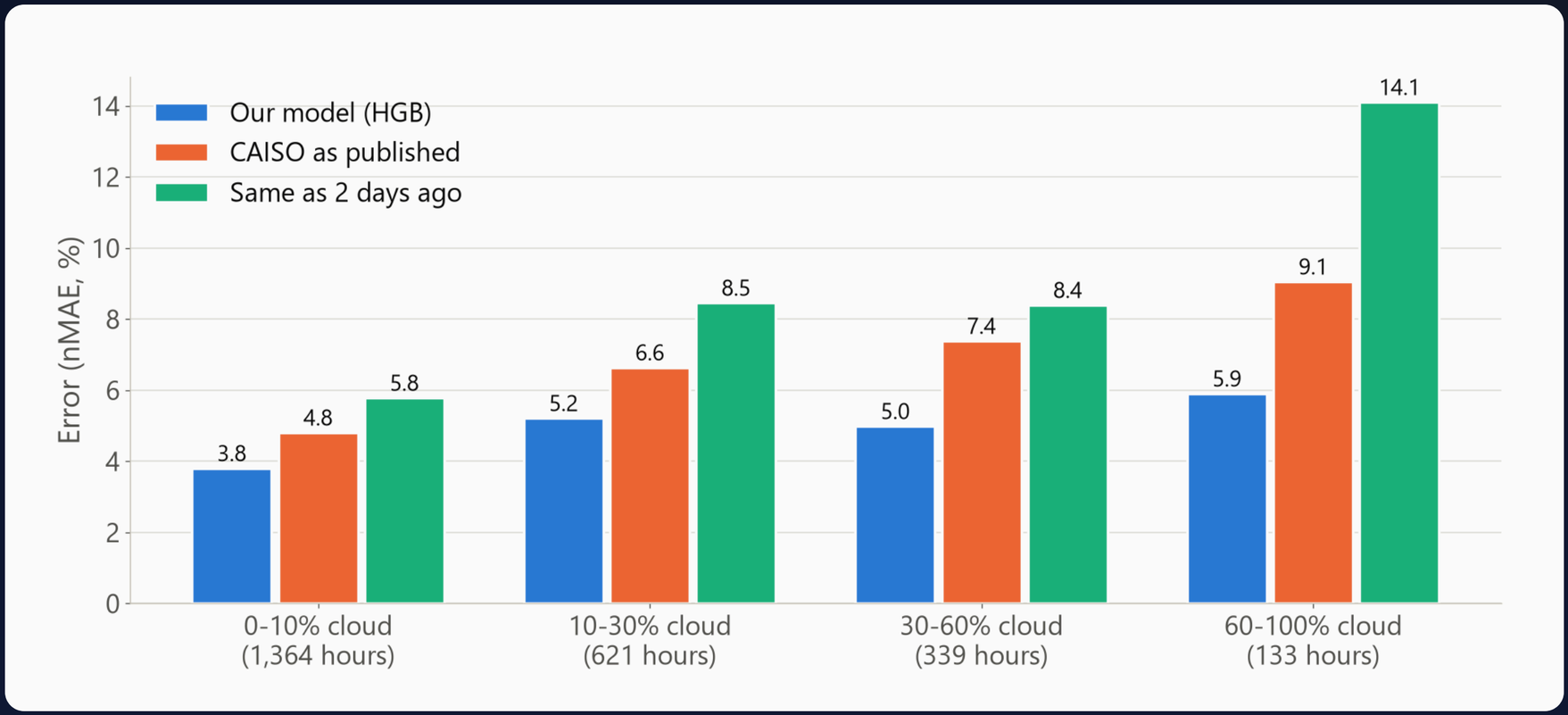
Five questions, five answers

OUR QUESTION	WHAT WE FOUND	VERDICT
1. Does the weather forecast help?	Our skill score is 0.40: our errors are 40% lower than the simple guess. We needed 0.30.	Supported
2. Close to CAISO's own forecast?	4.43% against a 7.32% limit: met. Against the adjusted limit (4.4294%) we are at 4.4348%: missed by 0.005 points, and we report it as a miss.	Partly supported
3. Flexible beats straight line?	4.43% for the tree model, 5.21% for the straight-line model.	Supported
4. Do clouds hurt?	5.9% error when very cloudy, 3.8% when clear.	Supported
5. Less backup power?	At the 95 of 100 hours level, 23% less than the simple guess. In practice runs it was 51%.	Supported

Questions and pass rules were saved before the test run (commit c242a1c). A "not supported" verdict would have stayed as it was.

RESULTS: WHERE THE ERROR LIVES

Clouds hurt every forecast. Ours holds up best.

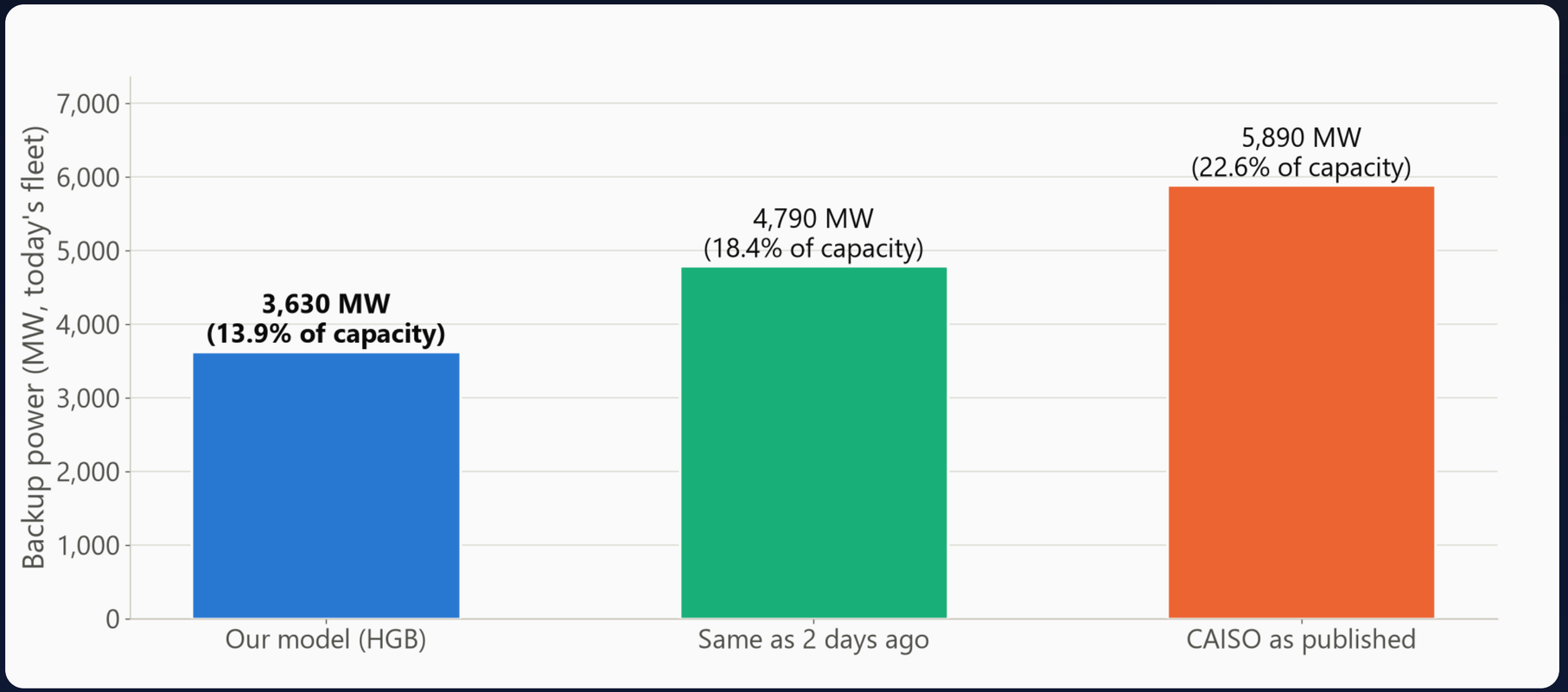


From clear to very cloudy, **our error rises 2.1 points, CAISO's 4.3, the simple guess 8.3.** Only 133 test hours were very cloudy, so treat the last group with care.

Error by the day-2 cloud forecast for each hour (known before the bid, not the observed sky). Daylight test hours, Apr 10 to Oct 17, 2026. Hours per group are under each label.

WHAT IT MEANS FOR THE GRID

A smaller error means less backup power to hold



To be covered in 95 of 100 daylight hours: **about 3,630 MW of backup with our forecast, 4,790 MW with the simple guess.** The error gap is worth about 3 TWh and $120 million a year: a size, not a savings promise.

Today's fleet: EIA 860M nameplate Sep 2026, all CISO, 26,034.5 MW. Hours when solar was turned down count as shortfalls here, which overstates CAISO's bar most. Price: $41 per MWh, CAISO 2024 average day-ahead.

LIMITS AND NEXT STEPS

What we learned, where it stops, what to do next

WHAT WE LEARNED

Free data gets close to the pros

A simple model on public data comes close to CAISO's own forecast and holds up best when it is cloudy.

LIMITS

Spring and summer only

The test has no winter. Spring 2026 had far more turned-down solar than spring 2025, so we forecast about 500 MW too high on average. History is 2.6 years.

WHAT WE RECOMMEND

Refit often, use as a second opinion

Refit every month, watch turned-down solar, use it next to CAISO's forecast, and add winter data as it arrives.

Winter was checked only in practice runs (Dec to Feb error 3.8% to 4.9%). Sources: EIA, CAISO, Open-Meteo.com (CC BY 4.0).